MIL-M-44312
9 December 1987

MILITARY SPECIFICATION

MILK, DRY, SUBSTITUTE, LOWFAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers spray dried, instantized milk product substitute which, when reconstituted with water, will yield a lowfat fluid milk product substitute of high beverage drinking qualities comparable to its commercial counterpart. The product is designed for use by the Department of Defense as a non-perishable item of limited issue.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, For Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8910

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

General Specifications for Approved Dairy Plants and Standards for Grades of Dairy Products (7 CFR Part 58)

Dairy Plants Surveyed and Approved for USDA Grading Service

(Application for copies should be addressed to the Dairy Division, Room 2750-S, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Food Safety and Inspection Service, Canning of Meat and Poultry Products (9 CFR Part 318)

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(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Bacteriological Analytical Manual for Foods

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington DC 20460.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists.

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

AMERICAN PUBLIC HEALTH ASSOCIATION

Standard Methods for the Examination of Dairy Products

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(Application for copies should be addressed to the American Public Health Association, 1015 18th Street, N.W. Washington, DC 20036.)

AMERICAN DAIRY PRODUCTS INSTITUTE

Standards for Grades of Dry Milks, Including Methods of Analysis; Bulletin 9

(Application for copies should be addressed to the American Dairy Products Institute, 130 N. Franklin Street, Chicago, IL 60606.)

ASSOCIATION OF VITAMIN CHEMISTS

Methods of Vitamin Assay

(Application for copies should be addressed to the Association of Vitamin Chemists, 8946 Major Avenue, Morton Grove, IL 60653.)

(Technical society and technical association documents are generally availab for reference from libraries. They are also distributed among technical group and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of th document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable law and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and fre from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Milk. The commingled milk used in the preparation of this product shall have a bacterial estimate of not more than 500,000/mL by SPC or DMC methods and a sediment test not to exceed 0.5 mg or equivalent. In addition, all other raw milk quality requirements and the handling of the milk shall be accordance with the provisions of the General Specifications for Approved Dair Plants and Standards for Grades of Dairy Products (7 CFR Part 58).

3.2.2 Lactose, edible. Edible lactose shall be white, free flowing, grind and shall comply with the Food Chemicals Codex (see 6.4.1).

3.2.3 Vegetable fat system. The following components and properties shall make up this blend so that the complete system contains 7.14 percent emulsifie and 0.05 percent antioxidant.

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a) Fat. Fat shall be a hydrogenated coconut fat (see 6.4.2) and shall conform to the following requirements:

Flavor and odor	Bland, no off flavors or odors
Moisture and impurities	0.10 percent maximum
Wiley melting point	91°F to 104°F
Solid Fat Index:	
50°F	54 to 60
70°F	30 to 36
80°F	6 to 10
92°F	3 maximum
Peroxide value	0.5 milliequivalent of peroxide per kg of fat
Free fatty acids	0.05 percent maximum
Stability (AOM)	200 hours (minimum)
Iodine Value	7.0 maximum
Color	1.5R/15.0 Y maximum

b) Emulsifier. Mono- and diglycerides (see 6.4.3) shall comply with the Food Chemicals Codex.

c) Antioxidant. The antioxidant shall be a propylene glycol/BHA/citric acid blend (see 6.4.4) and shall comply with the Food Chemicals Codex.

3.2.4 Silicon dioxide. Silicon dioxide anticaking agents (see 6.4.5) shall comply with the Food Chemicals Codex.

3.2.5 Lecithin, liquid. Liquid lecithin instantizing aid (see 6.4.6) shall comply with the Food Chemicals Codex.

3.2.6 Dioctyl sodium sulfosuccinate (DSS). Dioctyl sodium sulfosuccinate shall comply with the Food Chemicals Codex.

3.2.7 Vitamin A. Vitamin A shall be a concentrate of vitamin A ester palmitate. The palmitate shall be refined so that when added to the product at the required level, it will not impart a fishy or other objectionable flavor or odor to the product.

3.2.8 Vitamin D. Vitamin D shall be a concentrate of vitamin D food grade quality dissolved in an edible oil and shall meet flavor requirements of vitamin A.

3.2.9 Nitrogen gas. The nitrogen packaging gas shall be of food grade quality and may consist of pure nitrogen or a mixture of nitrogen and approximately 10 percent of carbon dioxide plus other inert gases in the atmosphere, but shall contain not more than 0.05 percent oxygen.

3.2.10 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

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3.3 Preparation and processing.

3.3.1 Preparation of skim milk. Skim milk shall be separated from raw milk meeting the requirements of 3.2.1.

3.3.2 Preparation of lowfat dry milk substitute.

3.3.2.1 Formulation. The lowfat milk shall be formulated so that the finished product complies with the requirements of 3.5 (see 6.4f). The lactose shall be used in such quantity that the finished product has no more than 2 percent lactose. The silicon dioxide, DSS and lecithin shall be incorporated in amounts in accordance with good manufacturing practice.

3.3.3 Processing.

a. Each step of the processing shall be performed in a continuous manner, protecting the product between steps from contamination or spoilage.

b. The concentrated skim milk and the fat system (see 3.2.3) containing the added vitamins A and D shall be preheated separately, then blended and thoroughly mixed.

c. The concentrated liquid lowfat milk substitute shall be standardized, homogenized, and pasteurized prior to drying.

d. Homogenization efficiency shall be considered acceptable if a minimum of 85 percent of the fat globules show a diameter of not more than 2 microns in size.

e. The product shall be dehydrated and instantized by spray drying, agglomeration, and redrying.

f. Alternatively, the product may be instantized by a single step spray drying process provided the finished product meets the requirements of 3.6.

g. Agglomeration shall be accomplished by using either a water moisturizing method, a steam method, or a method using water and steam in combination.

h. The dry instantized product shall be cooled to 90°F or 100°F following the final drying process.

i. The spray dryer shall be of the continuous removal type.

j. Wet collectors, if used, shall be USDA approved and the collected product shall not be recycled.

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k. If the instantizing process is to be performed at a location different from the original drying process, the spray dried lowfat milk substitute shall be kept sealed under a vacuum of 10 inches of mercury or under a nitrogen atmosphere , and held at a temperature of not more than 80°F. The instantizing shall be performed within 35 days from the initial drying process.

3.4 Can filling and sealing. Each can (see 5.1.1) shall be filled with product so as to conform to the finished product requirements and to the following requirements:

a. The instantized and cooled product shall be filled into the cans under an atmosphere of nitrogen so as to comply with the finished product headspace oxygen content requirement.

b. If not filled into the cans immediately after instantizing and cooling, the finished product shall be packaged in tightly sealed containers and held for not more than 24 hours at a temperature not to exceed 80°F.

3.5 Finished product requirements. The finished product shall conform to the following requirements:

a. There shall be no foreign material such as, but not limited to, glass, dirt, insect parts, hair, wood, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. The moisture content shall be not more than 3.00 percent.

e. The fat content shall be not less than 14.40 percent.

f. The protein content shall be not less than 29.0 percent.

g. The vitamin A content shall be not less than 8400 I.U. per 1 pound of product.

h. The vitamin D content shall be not less than 1700 I.U. per 1 pound of product.

i. The product density shall be not more than 0.40 g/cm^3.

j. The solubility index shall be not more than 1.00 mL.

k. The scorched particles shall be not more than Disc B (15.0 mg).

l. The bacterial estimate shall be not more than 30,000/g.

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m. The coliform count shall be not more than 10/g.

n. The salmonella test shall be negative.

o. The phosphatase test shall show not more than 1 microgram of phenol per mL of reconstituted milk.

p. The lot average headspace oxygen shall be not more than 3.0 percent with no individual unit sample being more than 4.5 percent.

Dry product

q. The dehydrated product shall have a white to light-cream color.

r. The dehydrated product shall be free flowing.

s. The dehydrated product shall be free from lumps.

t. No individual can shall contain less than 33.5 ounces of product.

u. The average net weight shall not be less than 34.0 ounces.

Reconstituted product.

v. Shall be fully reconstituted within 5 minutes with gentle spoon mixing.

w. Shall not develop more than 1/4 inch of foam on the surface after mixing and standing 2 minutes in an ordinary 8 ounce drinking glass

x. Shall have a natural milk color.

y. Shall have the flavor, aroma, and mouthfeel of fresh 2 percent lowfa milk.

z. Shall be free from lumps, undissolved particles, or grittiness.

3.5.1 Age of product. The finished product shall be delivered within 75 days after date of manufacturing and packaging. In the event two-step drying and instantizing is employed, this time period shall begin immediately followin the spray drying process.

3.6 Palatability. The finished product shall be equal to or better than the approved first article sample (see 4.4, 6.1, 6.3) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federa Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and shall comply with the

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General Specifications for Approved Dairy Plant and Standards for Grades of Dairy Products and be listed in the Publication of Dairy Plants Surveyed and Approved for USDA Grading Service.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of

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the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing and packing requirements. Any nonconformance revealed by act-ual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled can inspection. The inspection lot shall include only filled cans produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each can in the sample. The results shall be reported to the nearest 0.1 ounce. If the average net weight is less than 34.0 ounces or if the net weight of any individual can is less than 33.5 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The cans shall be selected at rand from the lot. The cans shall be opened and inspected for the defects listed in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejectio number
0 to 3200	8	-	0	2
	8	16	1	2
3201 to 35000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds th rejection number, the lot shall be rejected.

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c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equal or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product physical defects 1/ 2/ 3/

Category	Defect
Major	Dry product:
101	Not free flowing
102	Not free from lumps which do not fall apart under light finger pressure
103	Not white to light-cream color.
	Fully reconstituted product:
104	Not a natural milk color.
105	Product does not have the flavor, aroma, and mouthf of fresh 2 percent lowfat milk.
106	Not free from lumps, undissolved particles, or grittiness.
107	Product does not fully reconstitute within 5 minute with gentle spoon mixing.
108	Product develops more than 1/4 inch of foam on the surface after mixing and standing for 2 minutes in ordinary 8 ounce drinking glass.

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., moldy, rancid, sour, stale), foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved first article sample in palatability or overall appearance shall be cause for rejection of the lo (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ The product shall be fully reconstituted in accordance with the direction on the can label.

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4.5.3.3 Product analytical inspection. Eight randomly selected filled and sealed cans shall be opened and inspected for defects listed in table III. The lot size shall be expressed in cans. The sample unit shall be the contents of one can. The product shall be rejected if one or more of the defects in table III are present.

TABLE III. Product analytical and microbiological defects.

Category	Defect
Major	
101	Moisture content more than 3.00 percent.
102	Fat content less than 14.40 percent.
103	Protein content less than 29.0 percent.
104	Vitamin A (per 1 pound of product) less than 8400 I.U.
105	Vitamin D (per 1 pound of product) less than 1700 I.U.
106	Product density more than 0.40 g/cm^3.
107	Solubility index more than 1.00 mL.
108	Scorched particles more than Disc B (15.0 mg).
109	Phosphatase test result showing more than 1 microgram of phenol per mL of reconstituted milk.
110	The lot average headspace oxygen more than 3.0 percent.
111	Individual sample can has headspace oxygen more than 4.5 percent.

4.5.3.4 Product microbiological inspection. Twenty filled and sealed cans of finished product shall be selected at random from the lot. The contents of each sample shall be tested for salmonella and standard plate count in accordance with the Bacteriological Analytical Manual for Foods; Chapter: Detection and Identification of Salmonella and the Standard Methods for the Examination of

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Dairy Products; Chapter: Microbiological Count Methods, Section: Standard Plate Count. Any test failure shall be a critical defect and cause for rejection of the lot. Any of the following results shall be considered a test failure.

a. Salmonella test is positive.
b. Standard plate count moe than 30,000 bacteria/g.

4.5.4 Can condition examination. Examination of filled and sealed cans shall be in accordance with the United States Standards for Condition of Food Containers except that the inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4.1).

4.5.4.1 Can coating examination. The can coating shall be examined for defects in accordance with TT-C-495.

4.5.4.2 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

4.5.4.3 Can closure examination. Can closure shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and CFR 21, Part 113. Any nonconformance based on observation of can seam teardowns or of record of can seam teardowns shall be cause for rejection of any involved product.

4.5.4.4 Can leakage inspection. Can shall be inspected for leakage in accordance with 4.5.5.2. The sample unit shall be one filled and sealed can. The lot size shall be expressed in cans. The inspection level shallb e S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.5 Methods of inspection.

4.5.5.1 Chemical analysis. Chemical analysis shall be made in accordance with the methods from the publications referenced in table IV.

TABLE IV. Product tests

Characteristic	Source	Test method	Result reported
Fat	1/a	Fat in dried milk	0.01 percent
Protein	1/a	Protein (%N x 6.38)	0.1 percent
Moisture	2/	Toluene distillation	0.01 percent
Solubility index	2/	Solubility	0.01 mL
Scorched particles	2/	Scorched particles	0.1 mg
Oxygen in headspace	2/	Oxygen	0.1 percent
Vitamin D	1/b	Vitamin D Liquid chromatography	50 IU/lb.

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TABLE IV. Product tests (cont'd)

Characteristic	Source	Test method	Result reported
Vitamin A	3/	Colorimetric	50 IU/lb.
Phosphatase	4/	Scharer Rapid	0.1 microgram of phenol per mL of reconstituted milk

1/ Official Methods of Analysis of the A.O.A.C.
a. Chapter: Dairy Products, Section: Dried Milk, Nonfat Dry Milk and Malted Milks.

b. Chapter: Vitamins and Other Nutrients, Section: Vitamin D.

2/ American Dairy Products Institute, Bulletin 916, Standards for Grades of Dry Milks including Methods of Analysis.

3/ Methods of Vitamin Assay of the Association of Vitamin Chemists.

4/ Standard Methods for the Examination of Dairy Products, Chapter: Phosphatase Methods, Section: Scharer Rapid Phosphatase Test.

4.5.5.2 Can leakage test. Can seams of the filled and closed cans shall be examined for leakage by submerging the can in water contained in a vacuum desiccator, Mead Tester, or equivalent device and drawing a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches of Hg.) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by air entrapped in the double seam are not considered as signs of leakage.

4.5.6 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

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5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A. A net weight of 34 ounces of the product shall be filled into an open-top style, round metal can, size 603 x 812, with a soldered or welded side seam and compound-lined, double-seamed ends and shall be hermetically sealed under an atmosphere of nitrogen. The can shall not leak when tested in accordance with 4.5.5.2. A negative tolerance of 0.5 ounce may be allowed in any can providing that the average net weight is not less than the specified net weight. The can shall be made throughout from not less than 0.25 pound per base box electrolytic tin plate. Alternatively, can ends may be fabricated from ECCS plate, enameled both inside and out. The can shall be coated overall outside with a coating conforming to type I, class 2 of TT-C-495.

5.1.2 Level C. Product shallbe preserved in accordance with 5.1.1 except that cans with or without commercial exterior coating shall be acceptable. Alternatively, cans may be made from 0.20 pound per base box electrolytic tin plate, or electrolytic chromium-coated steel normally used for this product, provided the can is coated with a commercial exterior coating.

5.2 Packing. Packing shall be level A or B, as specified (see 6.1).

5.2.1 Level A packing. Six cans of the product, preserved as specified in 5.1, shall be packed on end in a snug-fitting fiberboard shipping container, constructed, closed, and reinforced in accordance with style RSC, grade V2s of PPP-B-636. The cans shall be arranged three in length, two in width, and one in depth within the container. Reinforcement shall be limited to nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Six cans of the product, preserved as specified in 5.1, shall be packed on end in a snug-fitting fiberboard shipping container, constructed, closed, and reinforced in accordance with style RSC, grade V3c, V3s, or V4s of PPP-B-636. The cans shall be arranged three in length, two in width, and one in depth within the container. Reinforcement shall be limited to nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.4 Labeling and marking.

5.4.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. In addition, the following directions for use shall be included on the label:

Directions for use of product.

1. 6 to 8 hours or longer prior to serving, add the contents of one can (34 ounces) to 7-1/2 quarts of 50° to 60°F potable water.

2. Mix with wire whip or spoon until the powder is visually dissolved.

3. Let stand 5 minutes; restir with a wire whip or spoon.

4. Refrigerate product 6 to 8 hours or longer to 40°F before serving.

If necessary, product may be consumed upon completion of the mixing process. However, chilling 6 to 8 hours or longer enhances flavor.

CAUTION: If electric mixer is used, mix on slow speed only.

Approximate yield - 2 gallons

5.4.2 Shipping containers. Shipping containers shall be marked in accordanc with MIL-STD-129.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved first article samples (see 3.6.2 and 6.3).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected tc be encountered during shipment, handling, and storage of the specific item bein procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 7-16/MCO 4030.33A/DLAR 4145.7.

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Technical information and guidelines. This document is based on a product developed under contract to the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA. The requirements and processing techniques utilized in preparation of this product provide a dry, lowfat milk substitute with a projected shelf life of 12 months when stored at 65° to 85°F and 6 months at 100°F. This stability is stated in terms of flavor and reconstitutibility with relative freedom from caking, browning, oxidation, and chalkiness. It has been determined that the followng processing guides should be helpful in preparation of the desired finished product.

a. Preparation of skim milk concentrate. The skim milk is preheated to 180°F, held for 15 seconds, and flashed in a vacuum pan resulting in a concentrated skim milk with 45 percent (± 5 percent) total solids.

b. Blending and standardization. The fat system and vitamin A and D concentrate are blended, heated to 150°F, and mixed for 5 minutes to insure uniform dispersion of vitamins, emulsifier, and antioxidant. The preheated fat system is blended with the concentrated skim milk, which has also been preheated to 150°F, in a ratio to yield a finished product with the required fat and vitamin A and D content.

c. Homogenization and pasteurization. The condensed skim milk fat system mixture is pasteurized at a temperature and time to constitute legal pasteurization. The mixture is homogenized at a minimum of 150°F at a pressure of 3000/500 psig or its equivalent to result in an 85 percent homogenization efficiency or better. After homogenizing and pasteurizing the product is cooled to 40°F.

d. Spray drying. The product is spray dried on a continuous removal type dryer and cooled to 90°F upon removal. It should have a moisture content of approximately 3.6 percent, a density of 0.58 g/cm^3, a solubility index of 125 mL, and scorched particles no greater than disc B (15.0 g).

e. Agglomeration. The spray dried product is mixed with seed lactose and silicon dioxide (2.0 percent maximum and 0.125 percent respectively). Agglomeration is accomplished by water moisturizing, a steam method, or a method utilizing water and steam. The moisturizing water and steam should contain DSS and lecithin which will result in a final dry product with 0.02 percent and 0.05 percent respectively. The agglomerated product is dried to less than 3 percent moisture and cooled to 80°F before canning. In the event the alternative method of drying and instantizing is utilized (see 3.3.3h), the lactose, silicon dioxide, lecithin, and DSS will be required to be added in the wet milk phase of blending prior to spray drying.

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F. Formulation. The lowfat milk substitute is formulated as follows:

Ingredient	Percent (by weight)
Skim milk solids	79.280
Milk fat (estimated0	0.575
Coconut fat	13.836
Emulsifier	1.106
Antioxidant	0.008
Lactose	2.000
Silicon dioxide	0.125
DSS (dioctyl sodium sulfosuccinate)	0.020
Lecithin	0.050
Moisture	3.000

6.4.1 Lactose, edible. It has been found that edible lactose manufactured by Foremost Foods Company (or its equivalent) meets the requirements of 3.2.2 and performs satisfactorily in this product.

6.4.2 Fat, hydrogenated coconut. It has been found that Hydrol 92 of the Glidden-Durkee Company (or its equivalent) meets the requirements of 3.2.3a and performs satisfactorily in this product.

6.4.3 Mono- and diglyceride emulsifier. It has been found that Atmos 150, manufactured by Atlas Chemical Company (or its equivalent) meets the requirements of 3.2.3b and performs satisfactorily in this product.

6.4.4 Antioxidant. It has been found that tenax A, manufactured by Eastman Chemical Products, Inc. (or its equivalent) meets the requirements of 3.2.3c and performs satisfactorily in this product.

6.4.5 Silicon dioxide. It has been found that Syloid 244, manufactured by W.R. Grace and Company (or its equivalent) meets the requirements of 3.2.4 and performs satisfactorily in this product.

6.4.6 Lecithin, liquid. It has been found that Centrolene-S liquid lecithin, manufactured by The Central Soya Company (or its equivalent) meets the requirements of 3.2.5 and performs satisfactorily in this product.

6.5 Subject term (key word) listing.

Beverage powder
Lowfat
Milk
Spray dried

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6.6 Supersession data. This specification includes the requirements of LP/PD 10-79A Milk, dry, substitute, lowfat dated 11 June 1980.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activity:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8910-0444

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-44312	Milk, Dry, Substitute, Lowfat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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